Cryptography and Network Security

TEXT BOOKS:

1. Cryptography and Network Security - Principles and Practice: William Stallings, Pearson Education, 6th Edition.
2. Cryptography and Network Security: Atul Kahate, Mc Graw Hill, 3rd Edition.

SUGGESTED READING:

1. Cryptography and Network Security: C K Shyamala, N Harini, Dr T R Padmanabhan, Wiley India, 1st Edition.
2. Cryptography and Network Security: Forouzan Mukhopadhyay, Mc Graw Hill, 3rd Edition.
3. Information Security, Principles, and Practice: Mark Stamp, Wiley India.
4. Principles of Computer Security: WM. Arthur Conklin, Greg White, TMH.
5. Introduction to Network Security: Neal Krawetz, CENGAGE Learning.
6. Network Security and Cryptography: Bernard Menezes, CENGAGE Learning.

ONLINE RESOURCES:

1. https://onlinecourses.nptel.ac.in/noc21_cs16/

UNIT – 1

- **Security Concepts:**
  Introduction, The need for security, Security approaches, Principles of security, Types of Security attacks, Security services, Security Mechanisms, A model for Network Security.

- **Cryptography Concepts and Techniques:**
  Introduction, plain text and cipher text, substitution techniques, transposition techniques, encryption and decryption, symmetric and asymmetric key cryptography, steganography, key range and key size, possible types of attacks.

INTRODUCTION

- Security is the key concept in this information age.
- When firms use private networks they do not think much about security, only a few do.
- But Internet has changed the scenario in the world. Specially for people running business over the internet, requires “SECURITY” about their transactions.

IMPORTANCE

- Security provides privacy for your data. Means no other party view your data.

- People who depend a lot on data security are
  - Daily Routine Transactions on the network
  - E-Mail, a common routine
  - E-commerce, Banking
  - Super Markets to Jet Flight
  - Multimedia Message Services (MMS)

While Transmitting data what people think about security

a) Is my data secured ?

b) If a sensitive data is send over inter network, can anybody else watch it

c) Can anybody alter my web site display ?

d) How can customers do business on internet

e) Can anybody misuse the credit card details that are send by the customers?

Background

- Information Security requirements have changed in recent times
- traditionally provided by physical and administrative mechanisms
- computer use requires automated tools to protect files and other stored information
- use of networks and communications links requires measures to protect data during transmission

Definitions

- **Computer Security** - generic name for the collection of tools designed to protect data and to thwart hackers
- **Network Security** - measures to protect data during their transmission
- **Internet Security** - measures to protect data during their transmission over a collection of interconnected networks
- **Information security -** is concerned with protecting information from getting in to the hands of unauthorized parties

Aspects of Security

- consider 3 aspects of information security:
  - **security attack**
  - **security mechanism**
  - **security service**

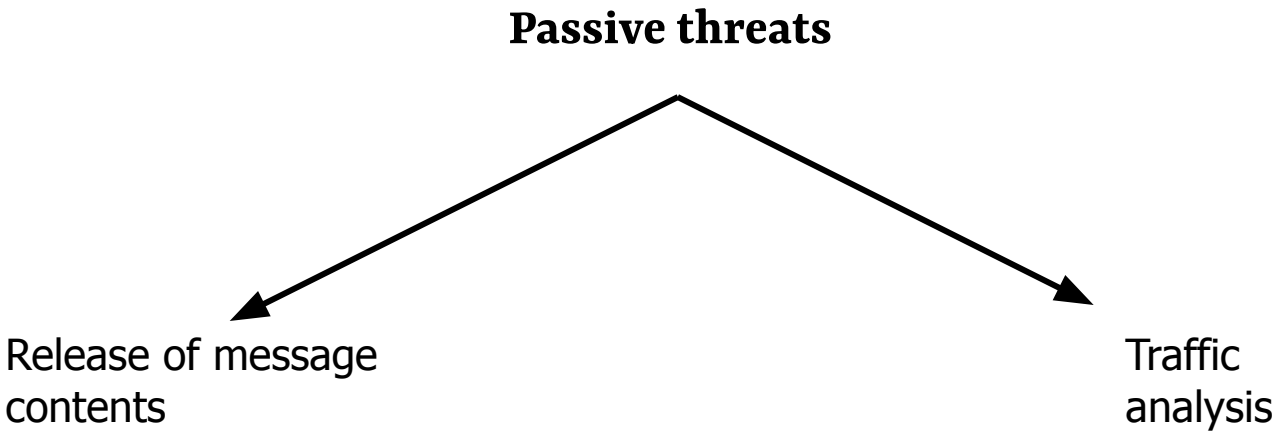
Services, Mechanisms, Attacks **(OSI Security Architecture)**

- Attack – action that compromises the security of information owned by an organization
- Mechanisms– detect, prevent or recover from a security attack
- Services – enhance the security of data processing systems and transfers – counter to security attacks

Security Attack

- any action that compromises the security of information owned by an organization
- information security is about how to prevent attacks, or failing that, to detect attacks on information-based systems
- often *threat* & *attack* used to mean same thing
- can focus of generic types or classification of attacks
  - passive
  - active

Passive Attacks



- eavesdropping, monitoring transmissions

Passive Attacks

- The goal of the opponent is to obtain information that is being transferred. Two types

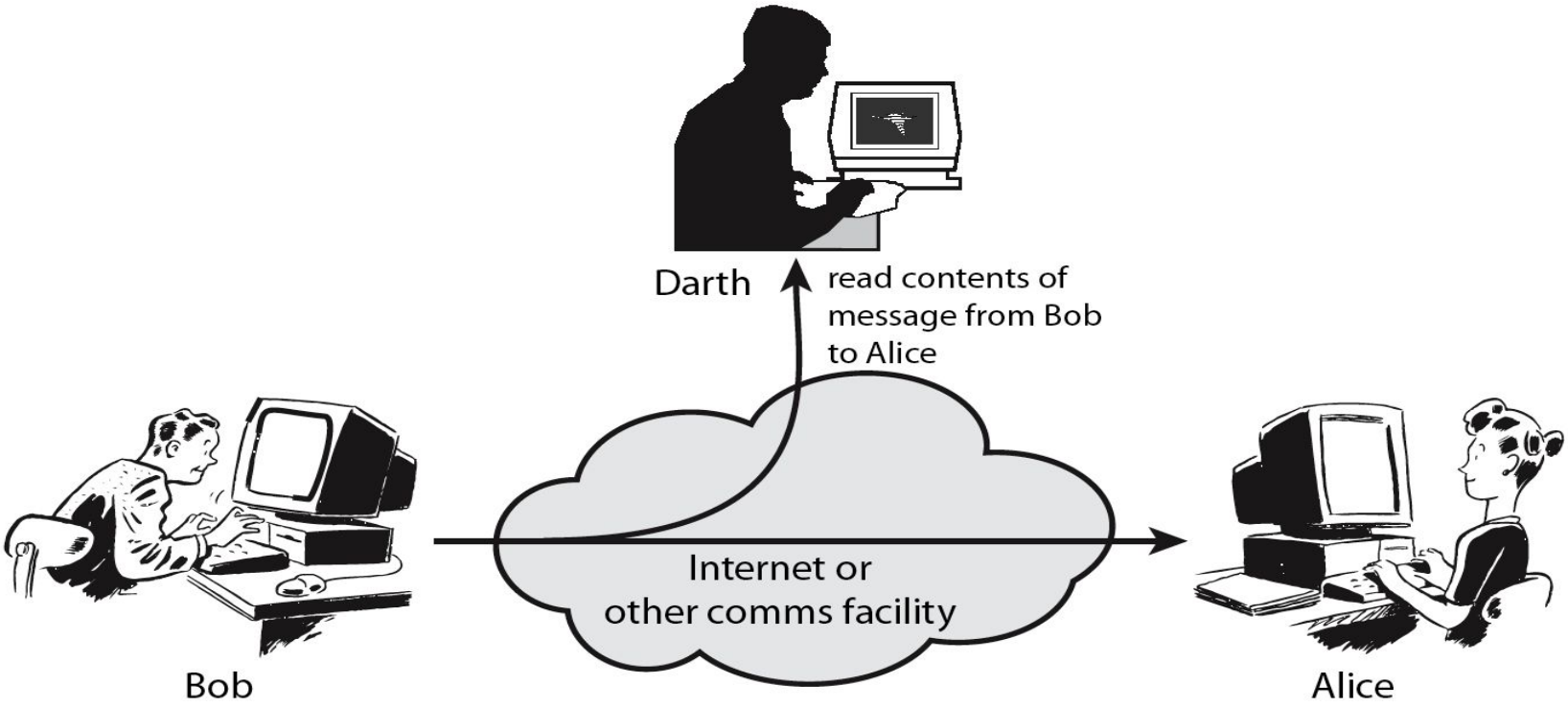
1) The release of message contents

- Ex: telephone conversation, an e-mail, a transferred file may contain sensitive or confidential information.

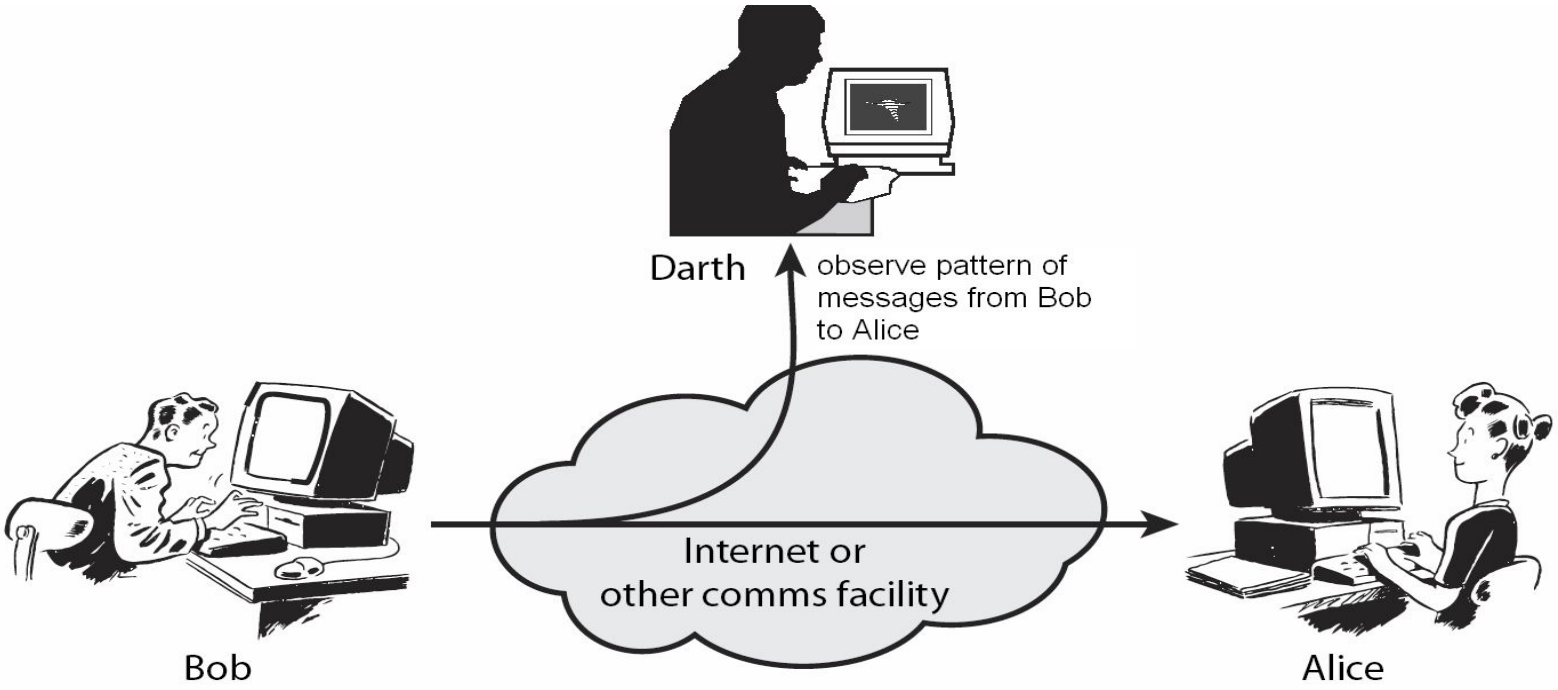
2) Traffic analysis

- A way of masking the contents of msgs or other information traffic so that opponents, even if they capture the msgs could not extract the information.
- The common technique for masking is encryption.
- If we had encryption protection place, an opponent might be able to observe the pattern of the msgs.
- The opponent could determine the location and identity of communicating hosts and could observe the frequency and length of msgs being exchanged.
- This information might be useful in guessing the nature of the communication that was taking place.

Passive Attacks

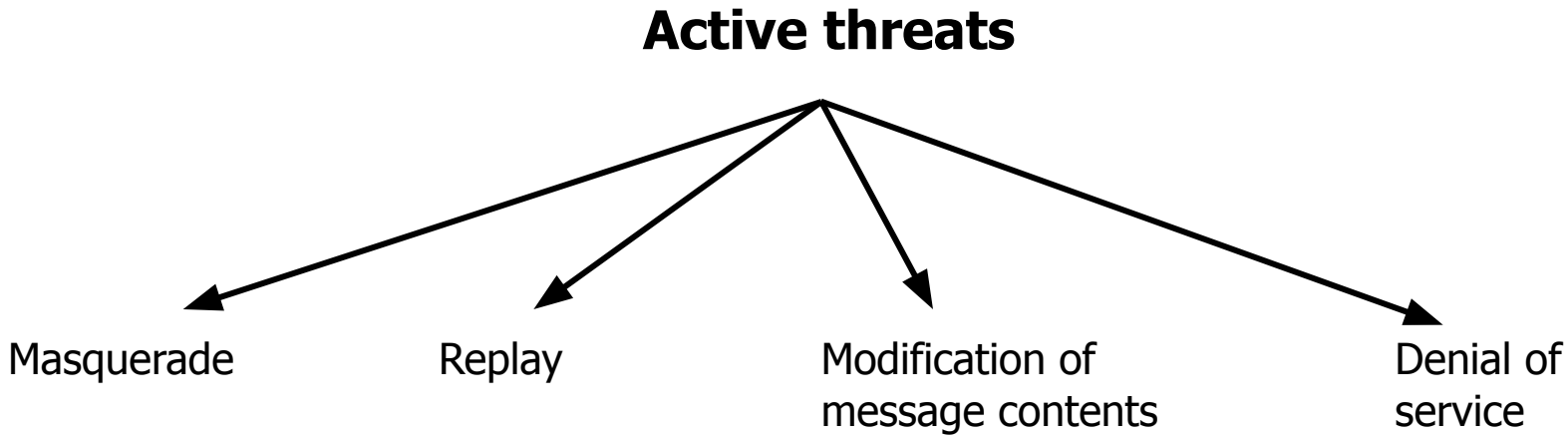


Release of message contents



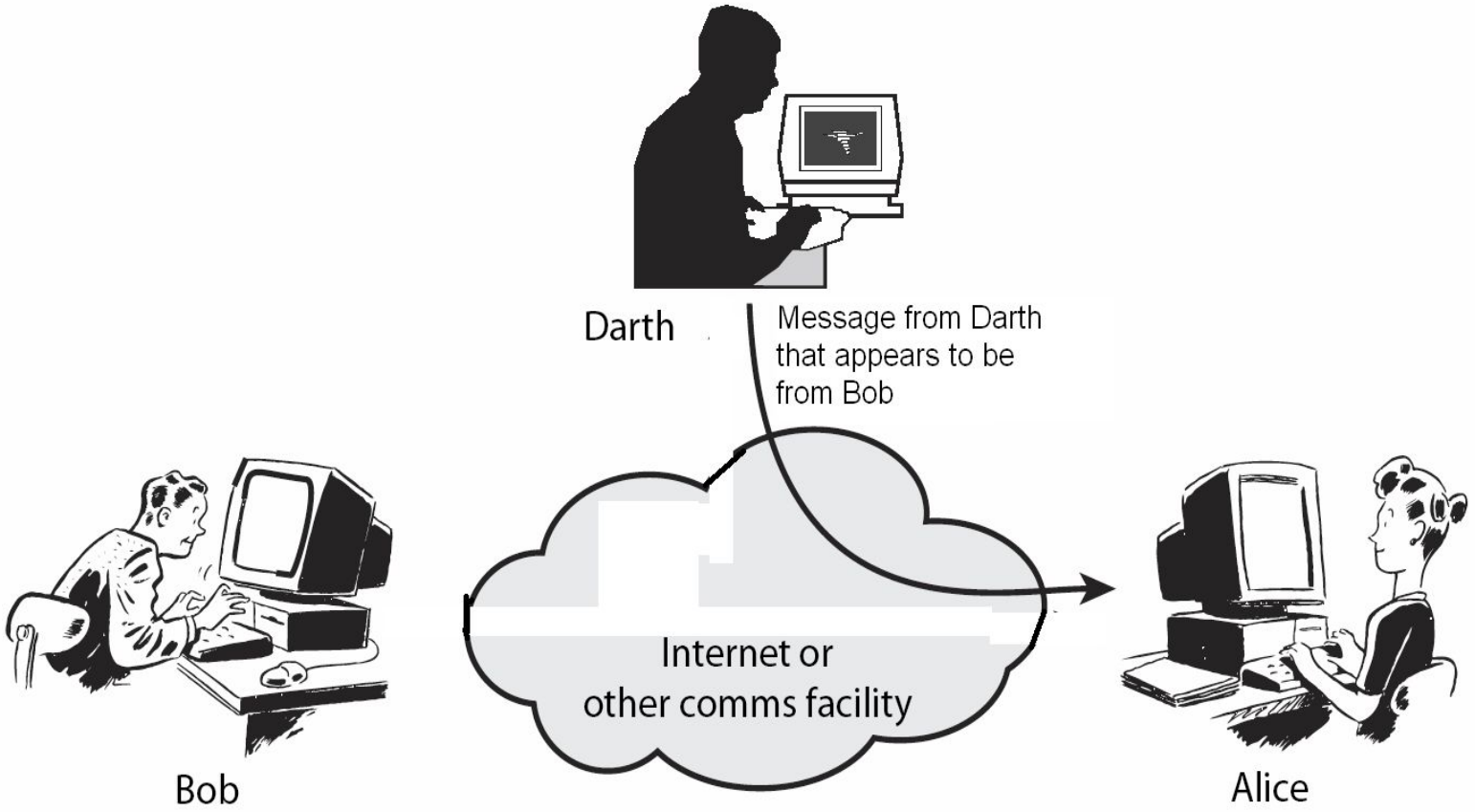
Traffic Analysis

Active Attacks



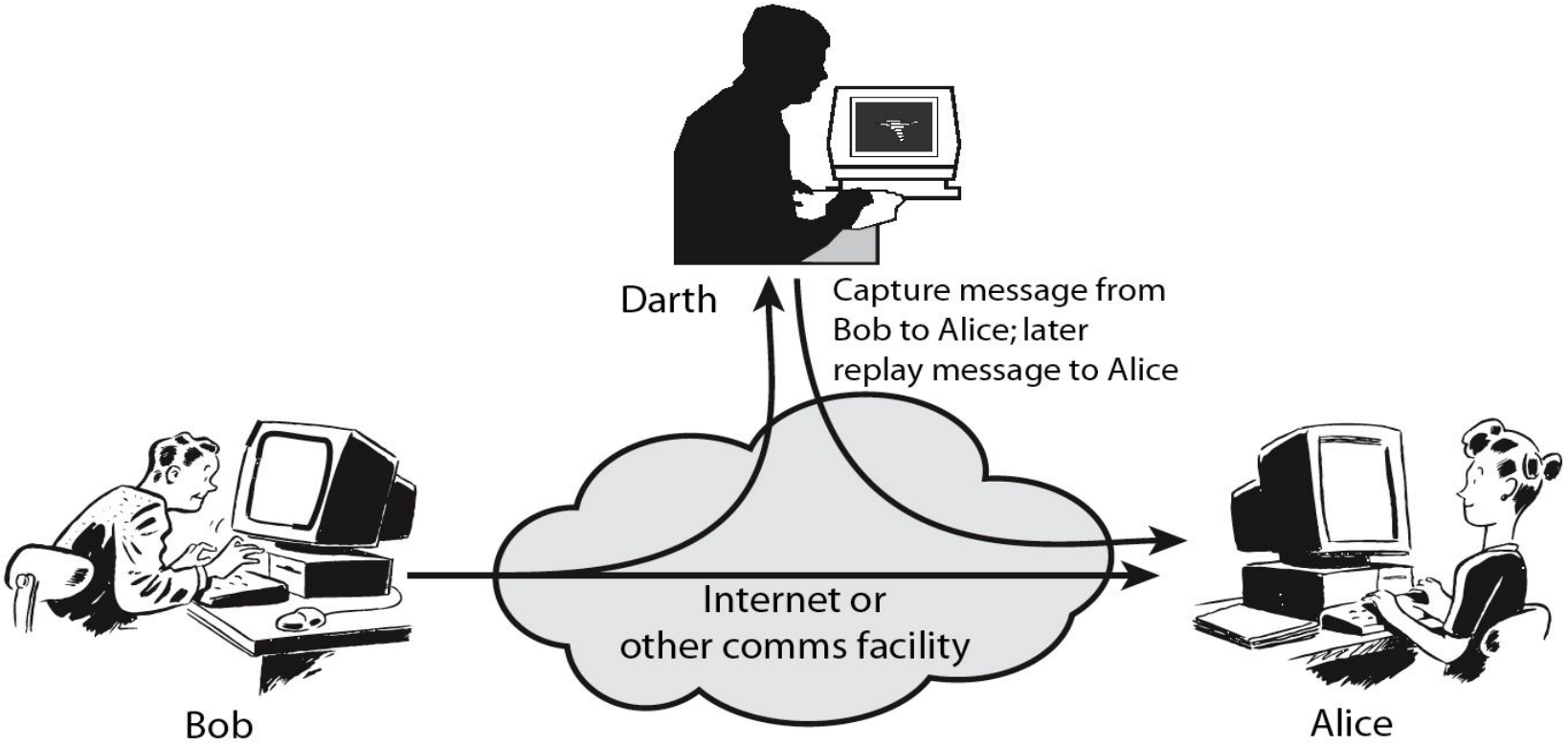
- some modification of the data stream

- A masquerade takes place when one entity pretends to be a different entity.
- Replay involves the passive capture of a data unit and its subsequent retransmission to produce an unauthorized effect.
- Modification of messages simply means that some portion of a legitimate msg is altered, delayed or reordered.
- Denial of service prevents or inhibits the normal use or management of communications facilities.

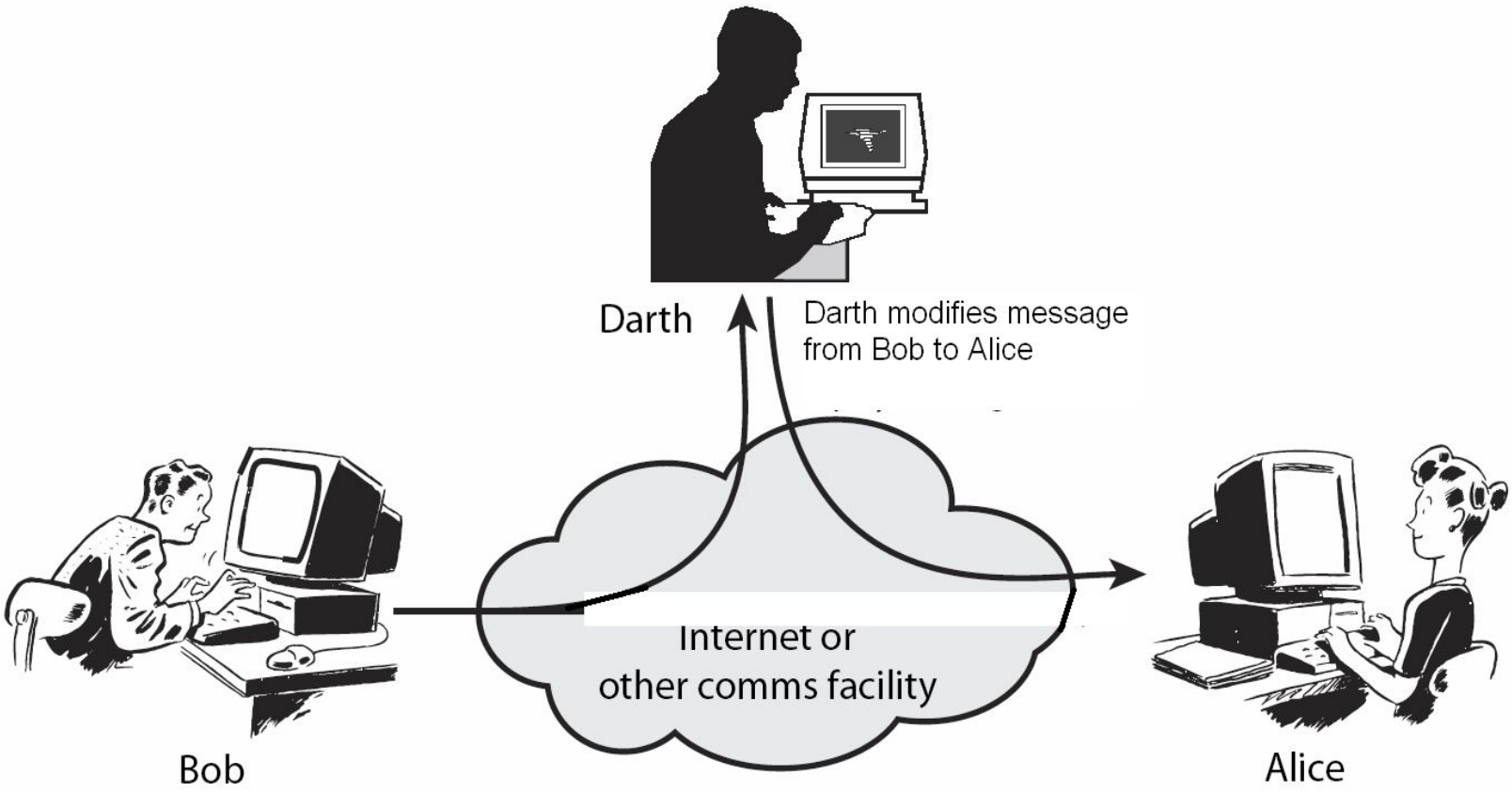

Darth
Message from Darth
that appears to be
from Bob
Internet or
other comms facility
Bob
Alice

Masquerade

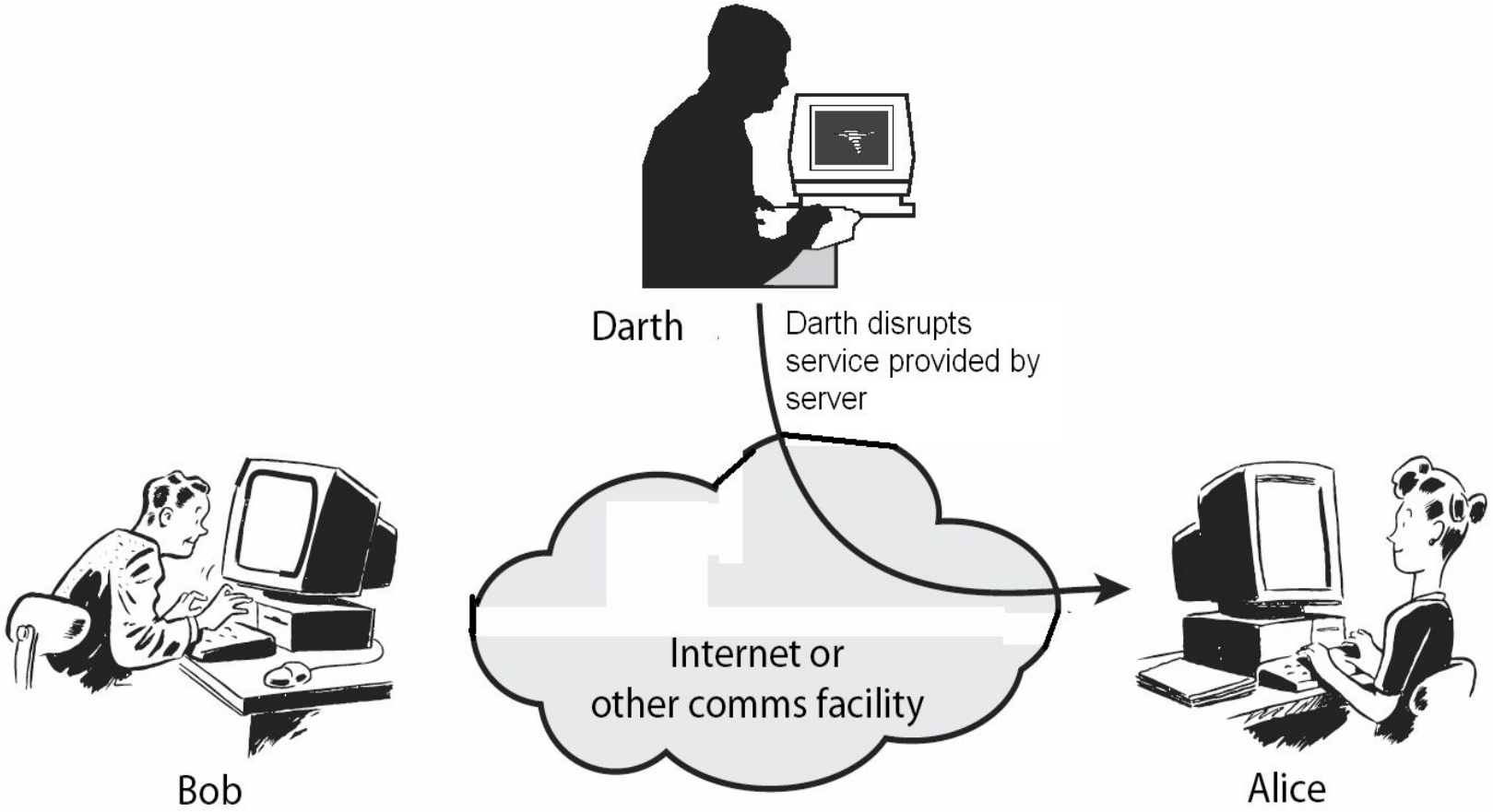
Active Attacks



Replay



Modification of messages



Denial of service

- Passive attacks are very difficult to detect because they do not involve any alteration of the data. However it is feasible to prevent the success of these attacks. Thus the emphasis in dealing with passive attacks is prevention rather than detection.

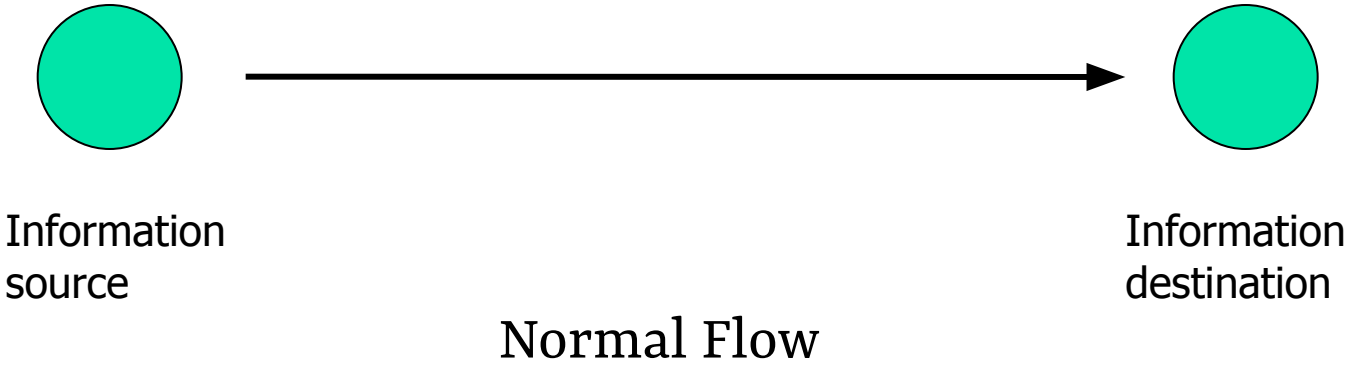
- Active attacks present the opposite characteristics of passive attacks. It is quite difficult to prevent because to do so would require physical protection of all communications facilities and paths at all times. Instead, the goal is to detect them and to recover. Because the detection has a deterrent effect, it may also contribute to prevention.

Categories of Security Attacks

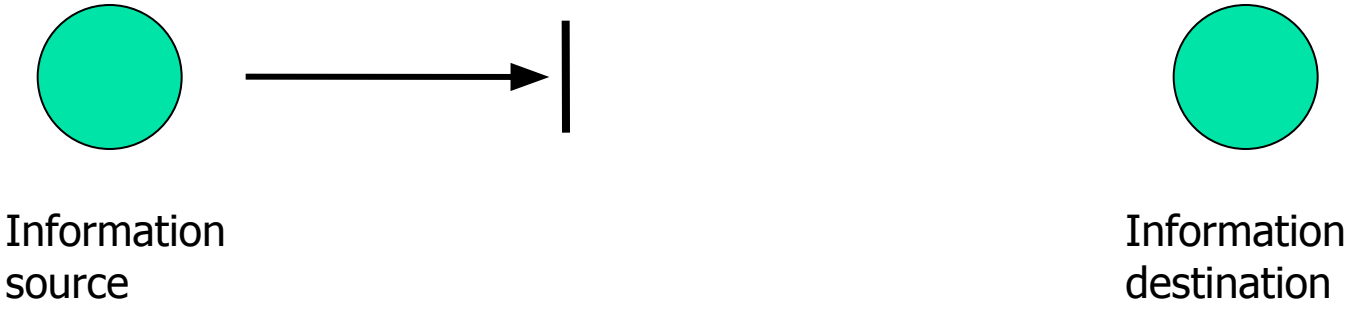
There are four categories of attacks

1) Interruption
2) Interception
3) Modification
4) Fabrication

Categories of Security Attacks



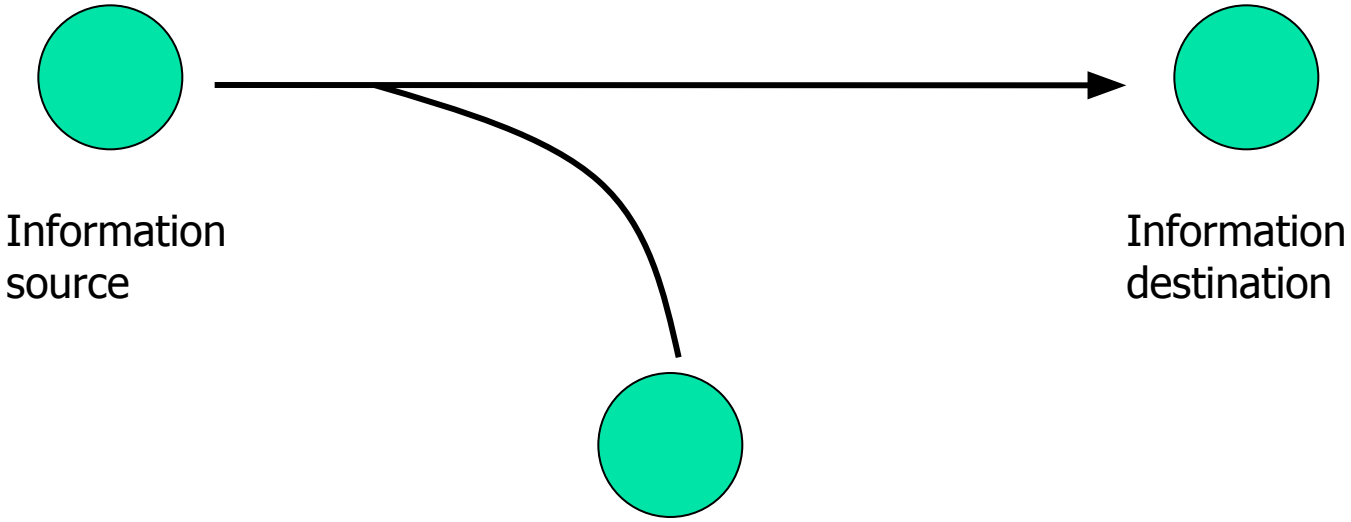
Security Attacks



Interruption

- Attack on **availability**

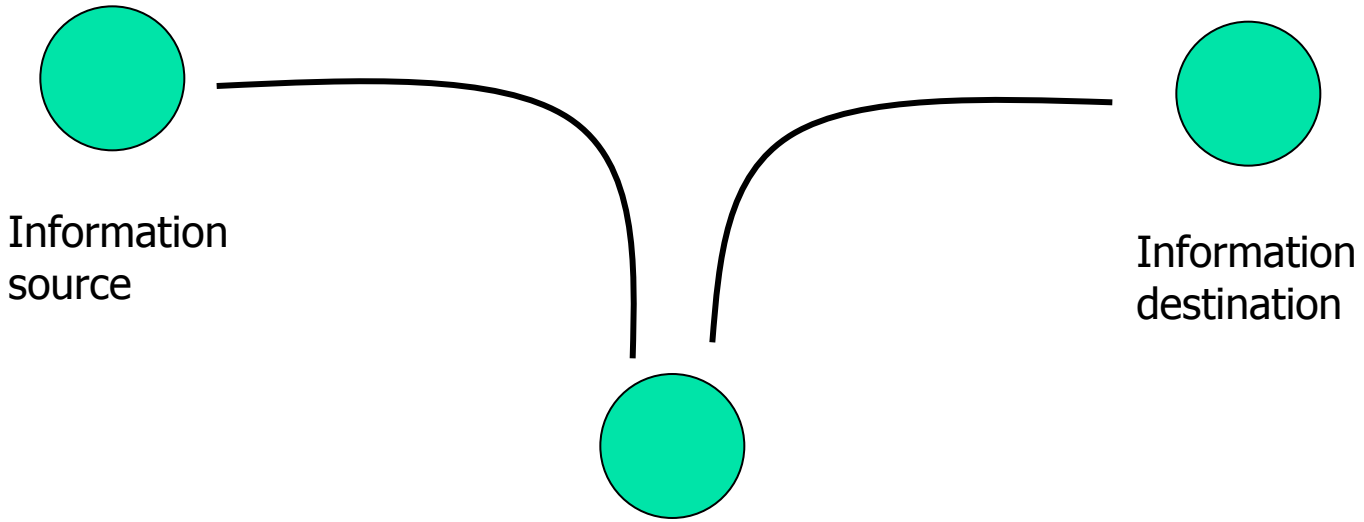
Ex: distruction pieces of h/w, such as hard disk, the cutting of communication line, disabling the file management system



Interception

- Attack on **confidentiality**

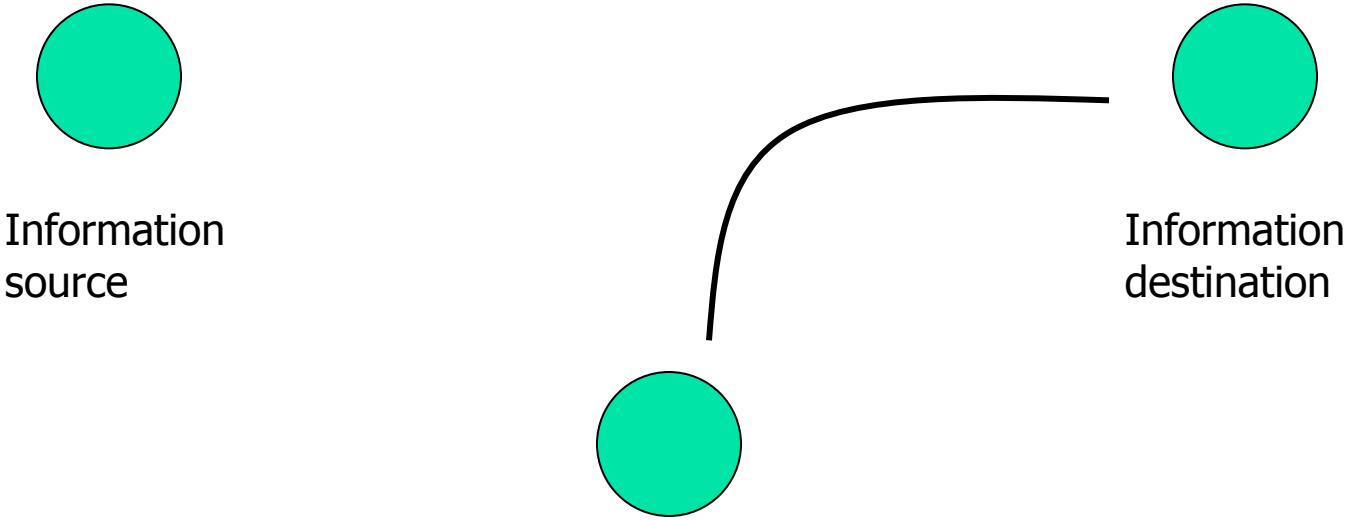
Ex: wiretapping to capture data in a n/w and the unauthorized copying of files or programs.



Modification

- Attack on **integrity**

Ex: changing values in a data file, altering a program, modifying



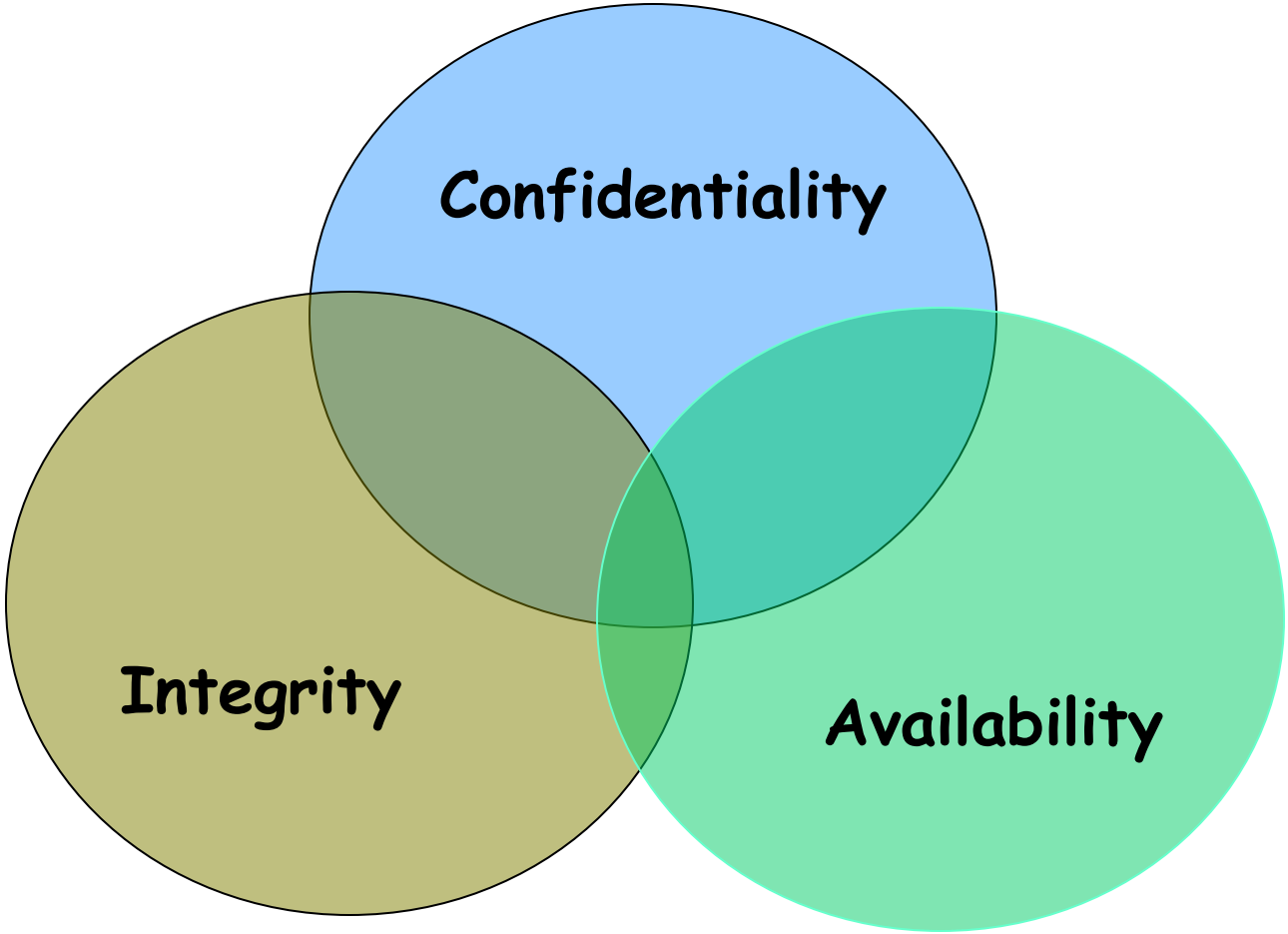
Fabrication

- Attack on **authenticity**

Ex: the insertion of spurious messages in a n/w or the addition of records to a file

Attack	Category	Attack on Service
Interruption	Active Attack	Availability
Interception	Passive Attack	Confidentiality
Modification	Active Attack	Integrity
Fabrication	Active Attack	Authentication

Security Goals



Security Service

- enhance security of data processing systems and information transfers of an organization
- intended to counter security attacks
- using one or more security mechanisms
- often replicates functions normally associated with physical documents
  - which, for example, have signatures, dates; need protection from disclosure, tampering, or destruction; be notarized or witnessed; be recorded or licensed

Security Services

1) Confidentiality – is the protection of transmitted data from passive attacks. protection of data from unauthorized disclosure.
2) Authentication – is concerned with assuring that a communication is authentic. assurance that the communicating entity is the one claimed.
3) Integrity – as with confidentiality, integrity can apply to a stream of msgs or selected fields within a msg. assurance that data received is as sent by an authorized entity.

32

4) Nonrepudiation – can't deny a message was sent or received.

5) Access Control – ability to limit and control access to host systems and applications. prevention of the unauthorized use of a resource.

6) Availability – attacks affecting loss or reduction on availability.

Security Mechanism

- feature designed to detect, prevent, or recover from a security attack
- no single mechanism that will support all services required
- however one particular element underlies many of the security mechanisms in use:
  - **cryptographic techniques**

Security Mechanisms (X.800)

- **specific security mechanisms:** may be incorporated into the protocol layer in order to provide some of the OSI security services
  - encipherment, digital signatures, access controls, data integrity, authentication exchange, traffic padding, routing control, notarization
- **pervasive security mechanisms:** not specific to any particular OSI security service or protocol layer.
  - trusted functionality, security labels, event detection, security audit trails, security recovery

SPECIFIC SECURITY MECHANISMS

- **Encipherment:** The use of mathematical algorithms to transform data into a form that is not readily intelligible.

- **Digital Signature:** Data appended to, or a cryptographic transformation of, a data unit that allows a recipient of the data unit to prove the source and integrity of the data unit and protect against forgery (e.g., by the recipient).

- **Access Control:** A variety of mechanisms that enforce access rights to resources.

- **Data Integrity:** A variety of mechanisms used to assure the integrity of a data unit or stream of data units.

- **Authentication exchange:** A mechanism intended to ensure the identity of an entity by means of information exchange.

- **Traffic Padding:** The insertion of bits into gaps in a data stream to frustrate traffic analysis attempts.

- **Routing Control:** Enables selection of particular physically secure routes for certain data and allows routing changes, especially when a breach of security is suspected.

- **Notarization:** The use of a trusted third party to assure certain properties of a data exchange.

PERVASIVE SECURITY MECHANISMS

Mechanisms those are not specific to any particular OSI security service or protocol layer.

- **Trusted Functionality:** That which is perceived to be correct w.r.t some criteria (e.g., as established by a security policy).

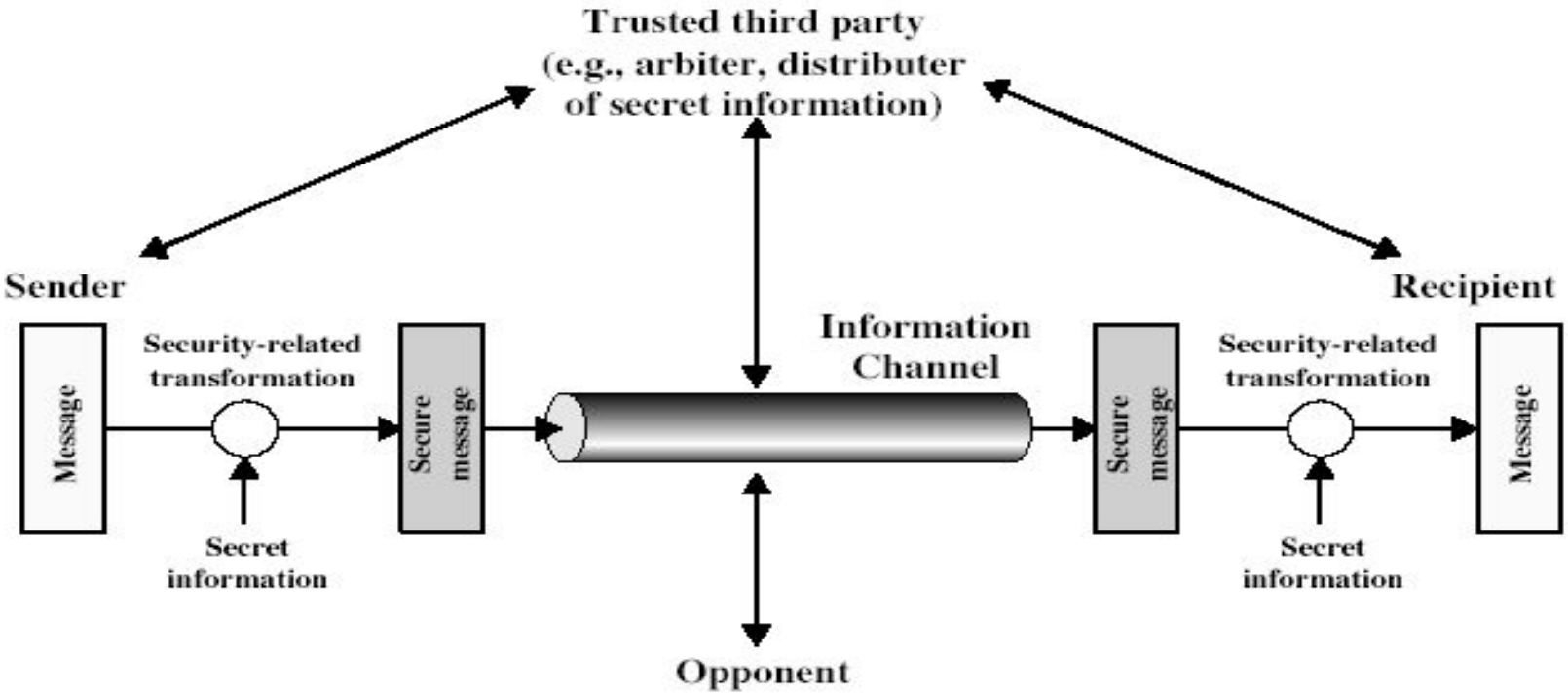
- **Security Label:** The marking bound to a resource (which may be a data unit) that names or designates the security attributes of that resource.

- **Event Detection:** Detection of security relevant events.

- **Security Audit Trail:** Data collected and potentially used to facilitate a security audit, which is an independent review and examination of system records and activities.

- **Security Recovery:** Deals with requests from mechanisms, such as event handling and management functions and takes recovery actions.

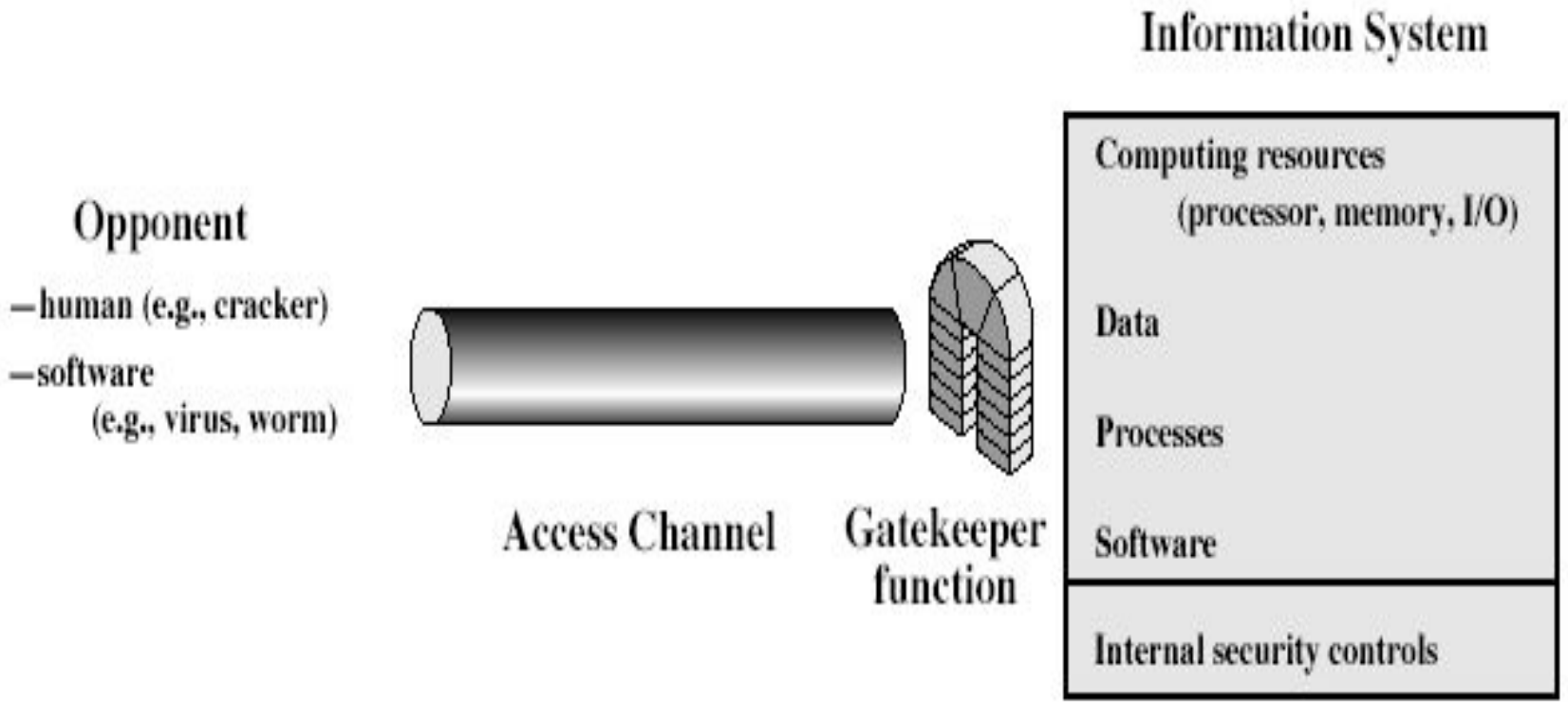
Model for Network Security



Model for Network Security

- using this model requires us to:
  1. design a suitable algorithm for the security transformation
  2. generate the secret information (keys) used by the algorithm
  3. develop methods to distribute and share the secret information
  4. specify a protocol enabling the principals to use the transformation and secret information for a security service

Model for Network Access Security



Model for Network Access Security

- using this model requires us to:
  1. select appropriate gatekeeper functions to identify users
  2. implement security controls to ensure only authorised users access designated information or resources
- trusted computer systems may be useful to help implement this model

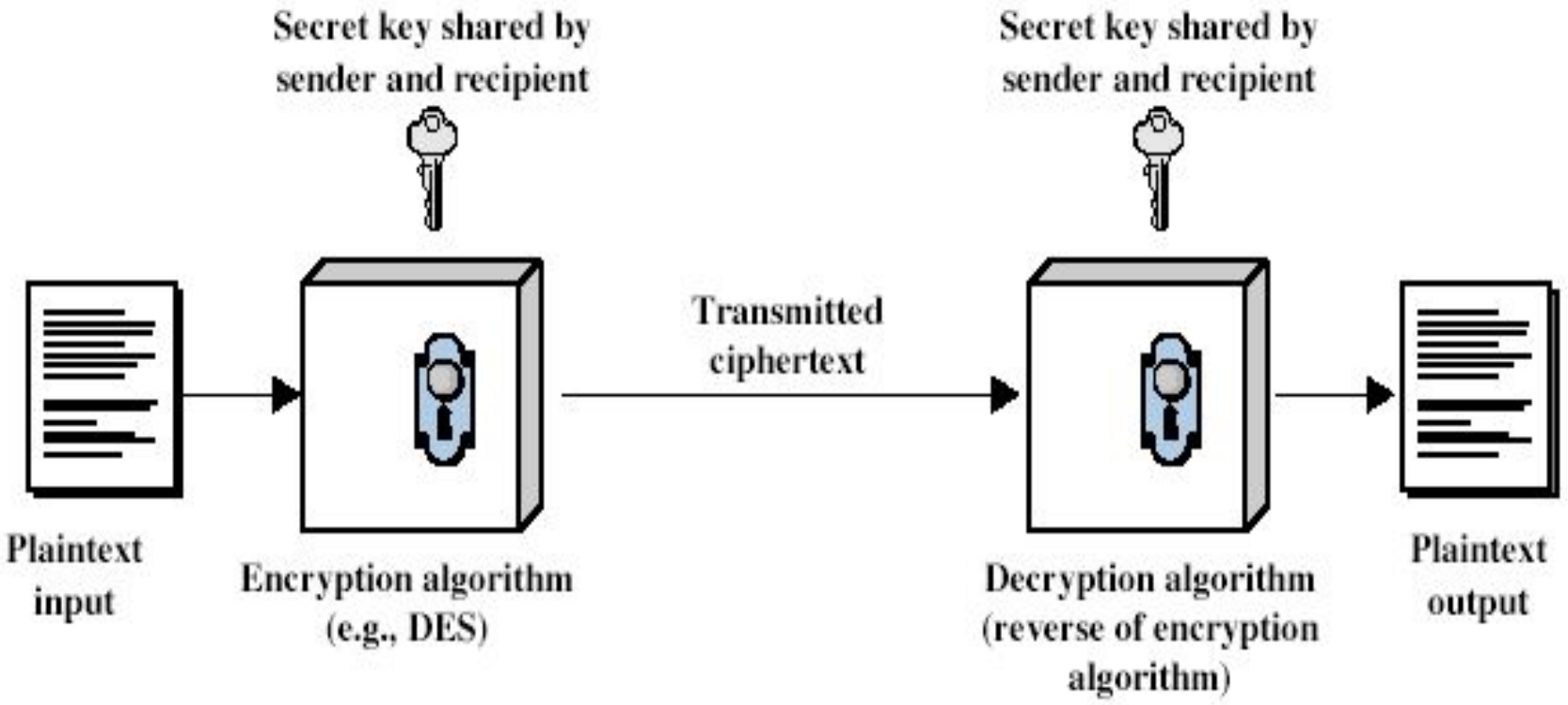
- Conventional encryption, also referred to as symmetric encryption, secret key encryption or single-key encryption
- was only type prior to invention of public-key in 1970’s

Conventional Encryption Principles

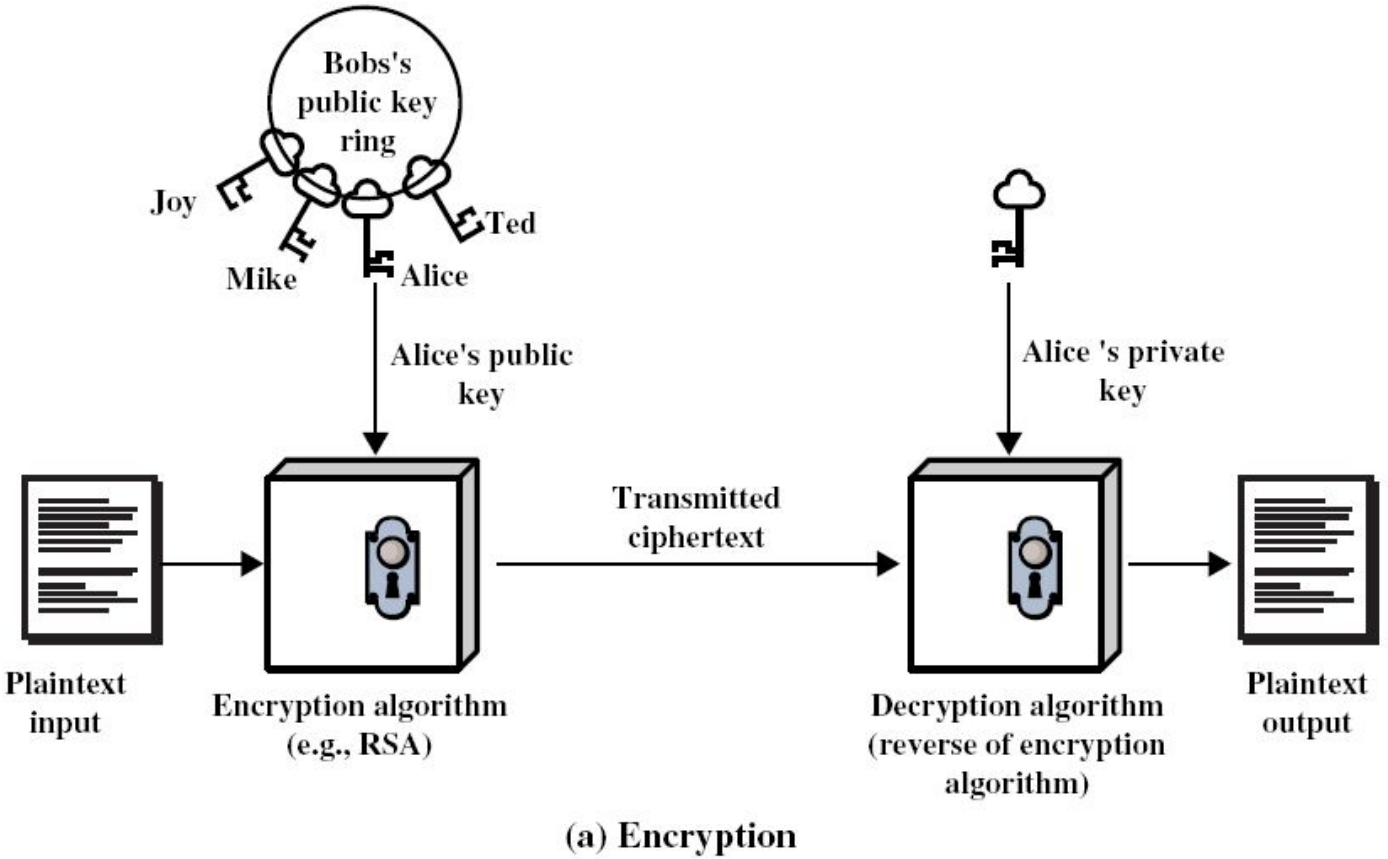
Has five ingredients

- **Plaintext**
- **Encryption algorithm**
- **Key**
- **Ciphertext**
- **Decryption algorithm**

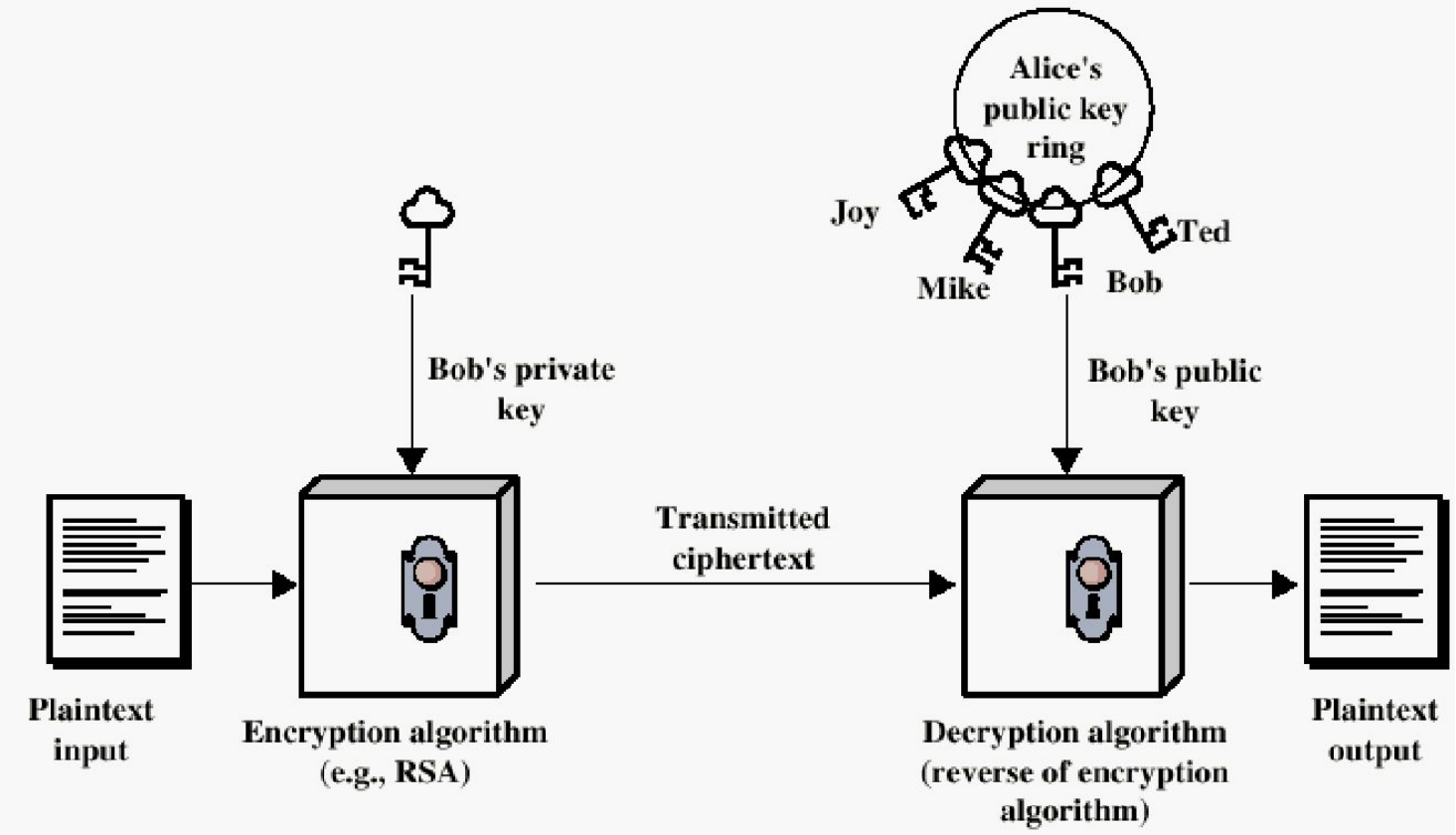
Symmetric Cipher Model



Asymmetric Cipher Model (for Encryption service)



Asymmetric Cipher Model (for Authentication service)



Basic Terminology

- **plaintext** - the original message
- **ciphertext** - the coded message
- **cipher** - algorithm for transforming plaintext to ciphertext
- **key** - info used in cipher known only to sender/receiver
- **encipher (encrypt)** - converting plaintext to ciphertext
- **decipher (decrypt)** - recovering ciphertext from plaintext
- **cryptography** - study of encryption principles/methods
- **cryptanalysis (codebreaking)** - the study of principles/ methods of deciphering ciphertext *without* knowing key
- **cryptology** - the field of both cryptography and cryptanalysis

Requirements

- two requirements for secure use of symmetric encryption:
  - We need a strong encryption algorithm
  - Sender and receiver must have obtained copies of the secret key in a secure fashion and must keep the key secure.

- assume encryption algorithm is known
- implies a secure channel to distribute key
- A source produces a message in plaintext, X = [X1, X2… XM] where M are the number of letters in the message.
- A key of the form K = [K1, K2… KJ] is generated. If the key is generated at the source, then it must be provided to the destination by means of some secure channel.
- With the message X and the encryption key K as input, the encryption algorithm forms the cipher text Y = [Y1, Y2, YN]. This can be expressed as
- Y = EK(X)
- The intended receiver, in possession of the k e y , is able to invert the
- transformation:
- X = DK(Y)
- An opponent, observing Y but not having access to K or X, may attempt to recover X or K or both. It is assumed that the opponent knows the encryption and decryption algorithms.
- If the opponent is interested in only this particular message, then the focus of effort is to recover X by generating a plaintext estimate. Often if the opponent is interested in being able to read future messages as well, in which case an attempt is made to recover K by generating an estimate.

Cryptography

- can characterize by:

1) The type of encryption operations used for transforming plaintext to ciphertext

- substitution / transposition / product

2) The number of keys used

- If both sender and receiver use the same key, the system is referred to as symmetric, single-key, secret key or conventional encryption. If different key, the system is referred to as asymmetric, two-key or public-key encryption.

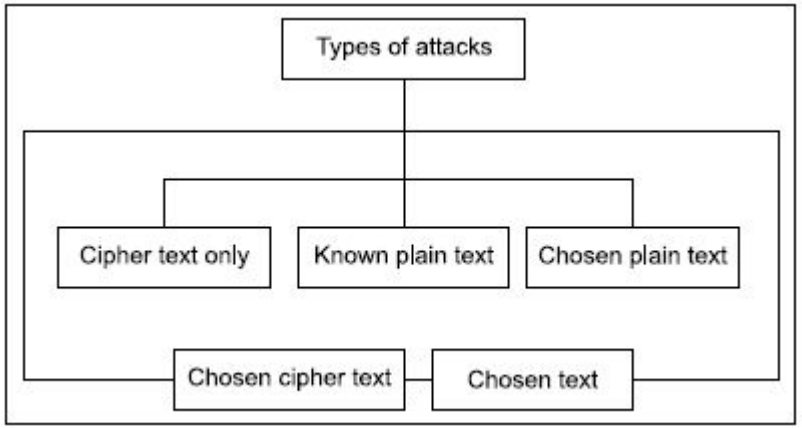
3) The way in which plaintext is processed

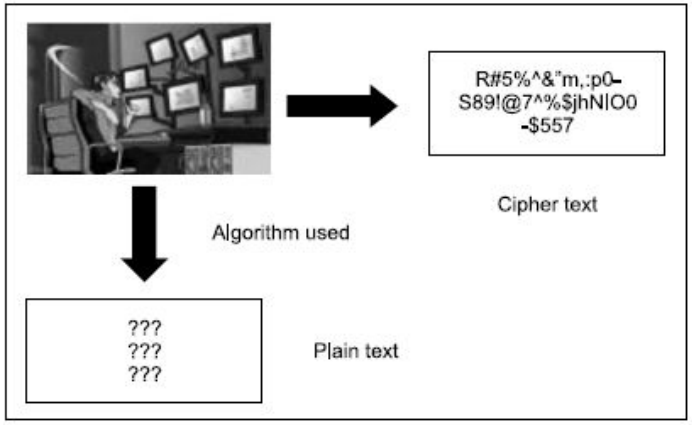
- block or stream cipher
- A block cipher processes the input one block of elements at a time, producing an output block for each input block.
- A stream cipher processes the input continuously, producing output one element at a time and so on.

Types of Cryptanalytic Attacks (Possible types of attacks)

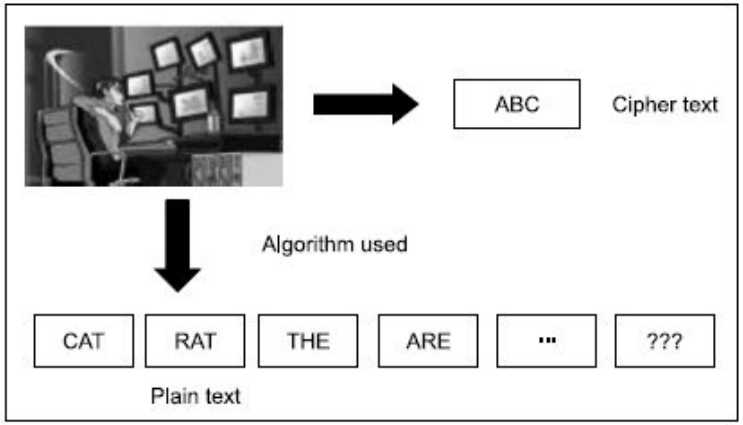
- **ciphertext only**
  - only known algorithm / ciphertext, statistical tests, can identify plaintext
- **known plaintext**
  - known/suspect plaintext & ciphertext to attack cipher
- **chosen plaintext**
  - select plaintext and obtain ciphertext to attack cipher
- **chosen ciphertext**
  - select ciphertext and obtain plaintext to attack cipher
- **chosen text**
  - select either plaintext or ciphertext to encrypt/decrypt to attack cipher

Types of Cryptanalytic Attacks

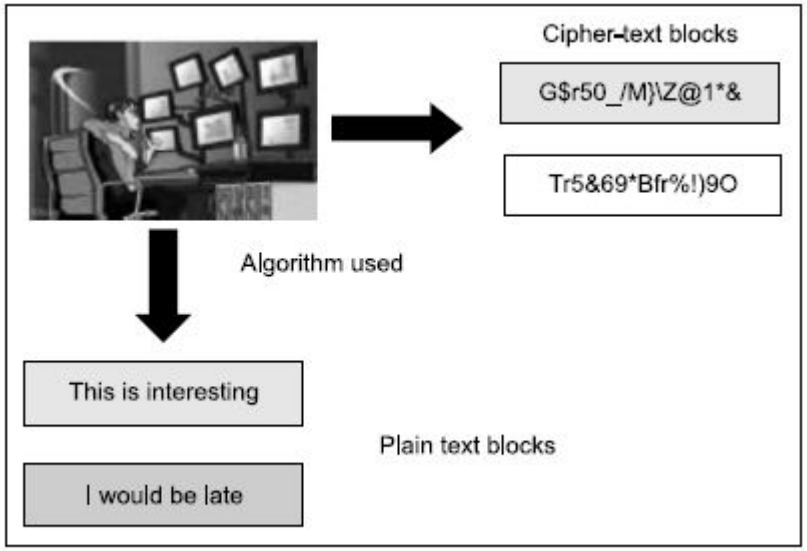




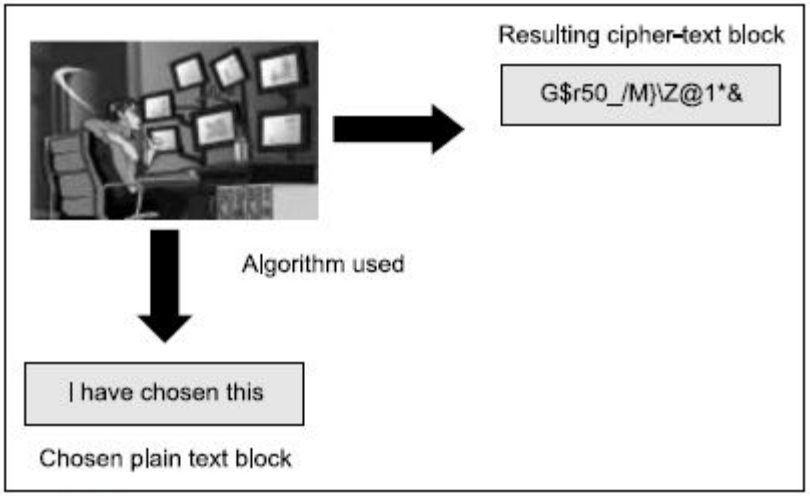
Cipher text- only attack



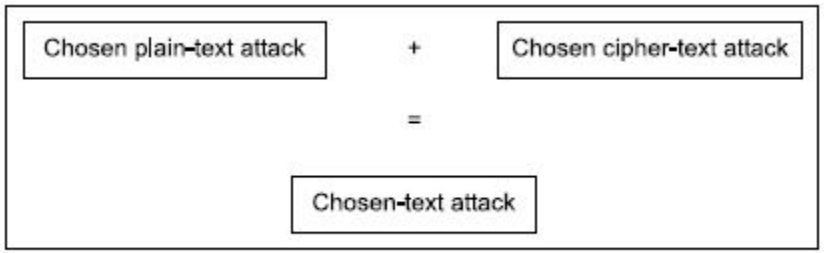
Confusion for the attacker



Known plain text attack



Chosen plain-text attack



Chosen text attack

Attack	*Things known to the attacker*	*Things the attacker wants to find out*
Cipher text only	• Cipher text of several messages, all of which are encrypted with the same encryption key • Algorithm used	• Plain-text messages corresponding to these cipher text messages • Key used for encryption
Known cipher text	• Cipher text of several messages, all of which are encrypted with the same encryption key • Plain-text messages corresponding to the above cipher text messages • Algorithm used	• Key used for encryption • Algorithm to decrypt cipher text with the same key
Chosen plain text	• Cipher text and associated plain-text messages • Chooses the plain-text to be encrypted	• Key used for encryption • Algorithm to decrypt cipher text with the same key
Chosen cipher text	• Cipher text of several messages to be decrypted • Corresponding plain-text messages	• Key used for encryption
Chosen text	• Some of the above	• Some of the above

ig. 2.74 Summary of types of attacks

Brute Force Search

- always possible to simply try every key
- most basic attack, proportional to key size
- assume either known / recognise plaintext

CLASSICAL ENCRYPTION TECHNIQUES

- There are two basic building blocks of all encryption techniques: substitution and transposition.

SUBSTITUTION TECHNIQUES

- A substitution technique is one in which the letters of plaintext are replaced by other letters or by numbers or symbols. If the plaintext is viewed as a sequence of bits, then substitution involves replacing plaintext bit patterns with cipher text bit patterns.

Caesar cipher (or) shift cipher

- The earliest known use of a substitution cipher and the simplest was by Julius Caesar.
- The Caesar cipher involves replacing each letter of the alphabet with the letter standing 3 places further down the alphabet.
- e.g., plain text : pay more money
- Cipher text: SDB PRUH PRQHB Note that the alphabet is wrapped around, so that letter following „z‟ is „a‟.
- For each plaintext letter p, substitute the cipher text letter c such that C = E(p) = (p+3) mod 26
- A shift may be any amount, so that general Caesar algorithm is

  C = E (p) = (p+k) mod 26 Where k takes on a value in the range 1 to 25.

  The decryption algorithm is simply P = D(C) = (C-k) mod 26

- letters of plaintext are replaced by other letters (or by numbers or symbols)
- earliest known example is Caesar Cipher, used by Julius Caesar in military communications
- replaced each letter by 3rd letter following it in the alphabet
- example:

```
meet me after the toga party
PHHW PH DIWHU WKH WRJD SDUWB
```

Modified Version of Caesar Cipher

- The Caesar cipher is good in theory, but not so good in practice.
- How can we generalize Caesar cipher a bit more?
- Let us assume that the cipher-text alphabets corresponding to the original plain-text alphabets may not necessarily be three places down the order, but instead, can be any places down the order. This can complicate matters a bit.
- Thus, we are now saying that an alphabet A in plain text would not necessarily be replaced by D. It can be replaced by any valid alphabet, i.e. by E or by F or by G, and so on. Once the replacement scheme is decided, it would be constant and will be used for all other alphabets in that message. An alphabet A can be replaced by any other alphabet in the English alphabet set, (i.e. B through Z). Of course, it does not make sense to replace an alphabet by itself (i.e. replacing A with A). Thus, for each alphabet, we have 25 possibilities of replacement.

Algorithm to break the modified Caesar cipher

1. Let *k* be a number equal to 1.
2. Read the complete cipher text message.
3. Replace each alphabet in the cipher text message with an alphabet that is *k* positions down the order.
4. Increment *k* by 1.
5. If *k* is less than 26, then go to Step 2. Otherwise, stop the process.
6. The original text message corresponding to the cipher-text message is one of the 25 possibilities produced by the above steps.

Cipher text	K	W	U	M		P	M	Z	M
Attempt Number (Value of *k*)									
1	L	X	V	N		Q	N	A	N
2	M	Y	W	O		R	O	B	O
3	N	Z	X	P		S	P	C	P
4	O	A	Y	Q		T	Q	D	Q
5	P	B	Z	R		U	R	E	R
6	Q	C	A	S		V	S	F	S
7	R	D	B	T		W	T	G	T
8	S	E	C	U		X	U	H	U
9	T	F	D	V		Y	V	I	V
10	U	G	E	W		Z	W	J	W
11	V	H	F	X		A	X	K	X
12	W	I	G	Y		B	Y	L	Y
13	X	J	H	Z		C	Z	M	Z
14	Y	K	I	A		D	A	N	A
15	Z	L	J	B		E	B	O	B
16	A	M	K	C		F	C	P	C
17	B	N	L	D		G	D	Q	D
18	C	O	M	E		H	E	R	E
19	D	P	N	F		I	F	S	F
20	E	Q	O	G		J	G	T	G
21	F	R	P	H		K	H	U	H
22	G	S	Q	I		L	I	V	I
23	H	T	R	J		M	J	W	J
24	I	U	S	K		N	K	X	K
25	J	V	T	L		O	L	Y	L

Fig. 2.13 Attempts to break modified Caesar-cipher text using multiple possibilities

Monoalphabetic Cipher

- shuffle (jumble) alphabetic arbitrarily
  - each plaintext letter maps to a ciphertext letter
- hence key is 26 letters long

```
Plain:  abcdefghijklmnopqrstuvwxyz
Cipher: DKVQFIBJWPESCXHTMYAUOLRGZN
Plaintext:  ifwewishtoreplaceletters
Ciphertext: WIRFRWAJUHYFTSDVFSFUUFYA
```

- Commonly used in puzzles (cryptoquotes in newspapers … not difficult to decode)

- Mono-alphabetic ciphers pose a difficult problem for a cryptanalyst because it can be very difficult to crack, thanks to the high number of possible permutations and combinations.
- There is only one hitch. If the cipher text created with this technique is short, the cryptanalyst can try different attacks based on his/her knowledge of the English language.
- As we know, some alphabets in the English language occur more frequently than others.
- Language analysts have found that given a single alphabet in cipher text, the probability that it is a P is 13.33%—the highest.
- After P comes Z, which is likely to occur 11.67%.
- The probability that the alphabet is C, K, L, N or R is almost 0—the lowest.

- Another way to improve on the simple monoalphabetic technique is to use different monoalphabetic substitutions as one proceeds through the plaintext message.
- The general name for this approach is polyalphabetic cipher. All the techniques have the following features in
- common.
- A set of related monoalphabetic substitution rules are used
- A key determines which particular rule is chosen for a given transformation.

Homophonic Substitution Cipher

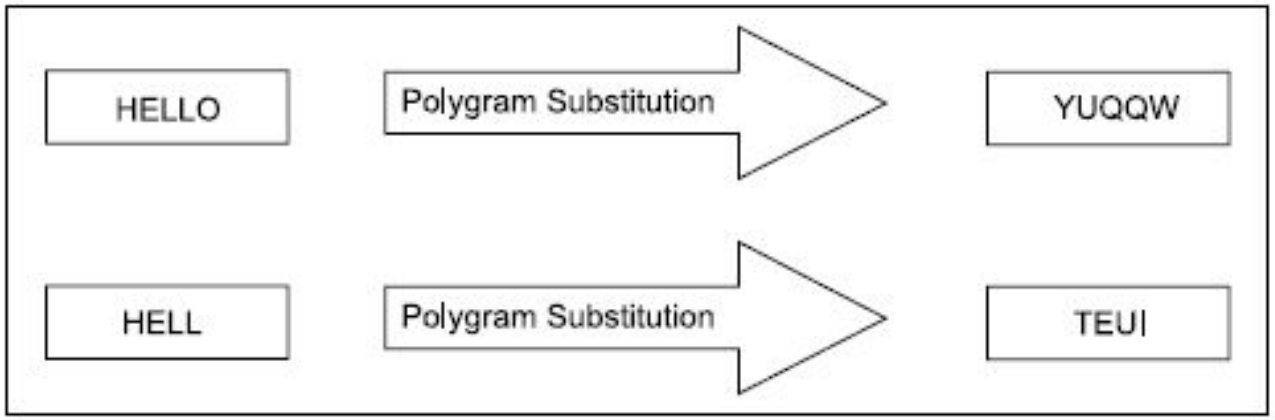
- The homophonic substitution cipher is very similar to mono-alphabetic cipher. Like a plain substitution cipher technique, we replace one alphabet with another in this scheme.
- However, the difference between the two techniques is that whereas the replacement alphabet set in case of the simple substitution techniques is fixed (e.g. replace A with D, B with E, etc.),
- in the case of homophonic substitution cipher, one plain-text alphabet can map to more than one cipher-text alphabet.
- For instance, A can be replaced by D, H, P, R; B can be replaced by E, I, Q, S, etc.
- Homophonic substitution cipher also involves substitution of one plain-text character with a cipher-text character at a time, however the cipher-text character can be any one of the chosen set.

Polygram Substitution Cipher

- In the polygram substitution cipher technique, rather than replacing one plain-text alphabet with one cipher text alphabet at a time, a block of alphabets is replaced with another block.
- For instance, HELLO could be replaced by YUQQW, but HELL could be replaced by a totally different cipher text block TEUI.
- This is true in spite of the first four characters of the two blocks of text (HELL) being the same.
- This shows that in the polygram substitution cipher, the replacement of plain text happens block by block, rather than character by character.

- Polygram substitution cipher technique replaces one block of plain text with another block ofcipher text—it does not work on a character-by-character basis.

Polygram Substitution Cipher



One-Time Pad (Vernam Cipher)

- Uses a truly random key as long as the message
  - the cipher will be secure, unbreakable
  - ciphertext bears no statistical relationship to the plaintext
- for **any plaintext** & **any ciphertext** (of equal length) there exists a key mapping one to the other
- a key can only be used **once**
- Problems: safe distribution of keys, running out of keys

- It is an unbreakable cryptosystem.
- It represents the message as a sequence of 0s and 1s.
- this can be accomplished by writing all numbers in binary, for example, or by using ASCII.
- The key is a random sequence of 0‟s and 1‟s of same length as the message. Once a key is used, it is discarded and never used again.
- The system can be expressed as Follows: Ci = Pi XOR Ki
- Ci - ith binary digit of cipher text
- Pi - i th binary digit of plaintext
- Ki - ith binary digit of key
- Thus the cipher text is generated by performing the bitwise XOR of the plaintext and the key. Decryption uses the same key. Because of the properties of XOR, decryption simply involves the same bitwise operation: Pi = Ci XOR Ki
- e.g., plaintext = 0 0 1 0 1 0 0 1 Key = 1 0 1 0 1 1 0 0
- ciphertext = 1 0 0 0 0 1 0 1

- Advantage: Encryption method is completely unbreakable for a ciphertext only attack.
- Disadvantages:

  It requires a very long key which is expensive to produce and expensive to transmit. Once a key is used, it is dangerous to reuse it for a second message; any knowledge on the first message would give knowledge of the second.

Playfair Cipher

The best-known multiple-letter encryption cipher is the Playfair, which treats digrams in the plaintext as single units and translates these units into cipher text digrams.

The Playfair algorithm is based on the use of a 5 x 5 matrix of letters constructed using a keyword.

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

In this case, the keyword is monarchy, and then filling in the remainder of the matrix with the remaining letters in alphabetic order

Plaintext is encrypted two letters at a time, according to the following rules:

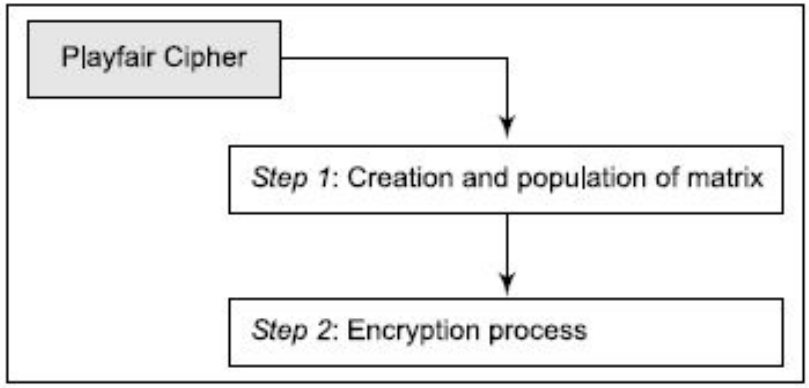
1.Two plaintext letters that fall in the same row of the matrix are each replaced by the letter to the right, with the first element of the row circularly following the last. For example, AR is encrypted as RM.

2.Two plaintext letters that fall in the same column are each replaced by the letter beneath, with the top element of the column circularly following the last. For example, MU is encrypted as CM.

3.Otherwise, each plaintext letter in a pair is replaced by the letter that lies in its own row and the column occupied by the other plaintext letter. Thus, HS becomes BP and EA becomes IM (or JM, as the encipherer wishes).

- Playfair cipher is a great advance over simple mono alphabetic ciphers. Since there are 26 letters, 26x26 = 676 diagrams are possible, so identification of individual diagram is more difficult.

Playfair Cipher steps

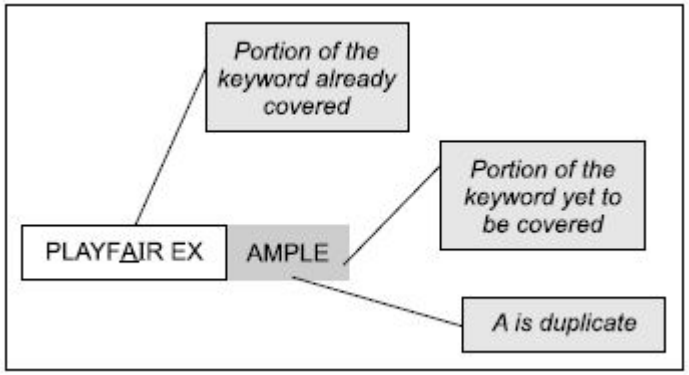


Playfair Cipher steps

Matrix Creation and Population

1. Enter the keyword in the matrix row-wise: left-to-right, and then top-to-bottom.
2. Drop duplicate letters.
3. Fill the remaining spaces in the matrix with the rest of the English alphabets (A-Z) that were not a part of our keyword. While doing so, combine I and J in the same cell of the table. In other words, if I or J is a part of the keyword, disregard both I and J while filling the remaining slots.

Situation when duplicate alphabet (for example A) is discovered



In such situation, disgard the repeating A and choose.

Play fair encryption process

1. Before executing these steps, the plain-text message that we want to encrypt needs to be broken down into groups of two alphabets. For example, if our message is MY NAME IS ATUL, it becomes MY NA ME IS AT UL. The encryption process works on this *broken-down* message.
2. If both alphabets are the same (or only one is left), add an X after the first alphabet. Encrypt the new pair and continue.
3. If both the alphabets in the pair appear in the same row of our matrix, replace them with alphabets to their immediate right respectively. If the original pair is on the right side of the row, then wrapping around to the left side of the row happens.
4. If both the alphabets in the pair appear in the same column of our matrix, replace them with alphabets immediately below them respectively. If the original pair is on the bottom side of the row, then wrapping around to the top side of the row happens.
5. If the alphabets are not in the same row or column, replace them with the alphabets in the same row respectively, but at the other pair of corners of the rectangle defined by the original pair. The order is quite significant here. The first encrypted alphabet of the pair is the one that is present on the same row as the first plain-text alphabet.

Decryption process works in the opposite direction.

Hill Cipher

- Multi-letter cipher
- Developed by Lester Hill in 1929
- Encrypts a group of letters: digraph, trigraph or polygraph

Prerequisites

Concepts to be known: Linear Algebra

For encryption process

- Matrix arithmetic modulo 26
- Square matrix

For decryption process

- Matrix arithmetic modulo 26
- Square matrix
- Determinant
- Multiplicative Inverse

Hill Cipher

- It is also a multi letter encryption cipher.
- It involves substitution of ‘m’ ciphertext letters for ‘m’ successive plaintext letters.
- For substitution purposes using ‘m’ linear equations, each of the characters are assigned a numerical values i.e. a=0, b=1, c=3, d=3,….. Z=25.
- For example if m=3, the system can be defined as:
- $c1= (k_{11}p_1+k_{12}p_2+k_{13}p_3)$ mod 26
- $c2=(k_{21}p_1+k_{22}p_2+k_{23}p_3)$mod 26
- $c3=(k_{31}p_1+k_{32}p_2+k_{33}p_3)$ mod26.
- If we represent in matrix form, the above statements as matrices and column vectors:

$$\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{pmatrix} \begin{pmatrix} p_1 \\ p_2 \\ p_3 \end{pmatrix} \text{ mod } 26$$

- Thus C = K P mod 26, where C= Column vectors of length 3

  P= column vectors of length 3

  K= 3X3 encryption key matrix.

For decryption process, inverse of matrix K i.e. K^{-1} is required which is defined by the equation $K\ K^{-1}=K^{-1}\ K=I$,
where I is the identity matrix that contains only 0's and 1's as its elements.

Plaintext is recovered by applying K^{-1} to the cipher text. It is expressed as
$C=E_k(P)=K\ P \bmod 26$

$P= D_k(C)=K^{-1}\ C \bmod 26=K^{-1}KP= I\ P= P$

Example: P= 'I can't do it'
Size of m is 3
I can't do it
8 2 0 13 19 3 14 8 19
and key K is chosen as following

$$\begin{pmatrix} 9 & 18 & 10 \\ 16 & 21 & 1 \\ 5 & 12 & 23 \end{pmatrix}$$

Hill cipher classical Encryption Technique ①

Encryption process: $C = KP \bmod 26$

Ex: plaintext = ATTACK

$$\text{Key} = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$$

Since key is a 2×2 matrix, plaintext should be converted into vectors of length 2 [square matrix]

So, $\begin{bmatrix} A \\ T \end{bmatrix} \begin{bmatrix} T \\ A \end{bmatrix} \begin{bmatrix} C \\ K \end{bmatrix}$

① Encryption process

1st vector $P = \begin{bmatrix} A \\ T \end{bmatrix} = \begin{bmatrix} 0 \\ 19 \end{bmatrix}$, $\text{Key} = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$

A = 0 ⋮ Z = 25

$$C = KP \bmod 26$$

$$= \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 0 \\ 19 \end{bmatrix} \bmod 26 = \begin{bmatrix} 2\times0 + 3\times19 \\ 3\times0 + 6\times19 \end{bmatrix} \bmod 26$$

$$C = \begin{bmatrix} 57 \\ 114 \end{bmatrix} \bmod 26 = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

Corresponds to letters $= \begin{bmatrix} F \\ K \end{bmatrix}$

∴ plain text $\begin{bmatrix} A \\ T \end{bmatrix}$ gives $\begin{bmatrix} F \\ K \end{bmatrix}$ ciphertext

AT → FK

(2) 2nd vector is $\begin{bmatrix} T \\ A \end{bmatrix} = \begin{bmatrix} 19 \\ 0 \end{bmatrix}$

$C = KP \bmod 26$

$= \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 19 \\ 0 \end{bmatrix} \bmod 26 = \begin{bmatrix} 2 \times 19 + 3 \times 0 \\ 3 \times 19 + 6 \times 0 \end{bmatrix} \bmod 26$

$C = \begin{bmatrix} 38 \\ 57 \end{bmatrix} \bmod 26 = \begin{bmatrix} 12 \\ 5 \end{bmatrix}$

$\begin{bmatrix} T \\ A \end{bmatrix} = \begin{bmatrix} M \\ F \end{bmatrix}$ $TA \rightarrow MF$

(3) $\begin{bmatrix} C \\ K \end{bmatrix} = \begin{bmatrix} 2 \\ 10 \end{bmatrix}$

$C = KP \bmod 26$

$= \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 10 \end{bmatrix} \bmod 26$

$C = \begin{bmatrix} 34 \\ 66 \end{bmatrix} \bmod 26 = \begin{bmatrix} 8 \\ 14 \end{bmatrix} = \begin{bmatrix} I \\ O \end{bmatrix}$

$CK \rightarrow IO$

Plaintext = ATTACK

FKMFIO

Hill cipher algorithm

Decryption process: $P = K^{-1} C \bmod 26$

Plaintext → ATTACK Key $K = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$

Ciphertext → FKMFIO

① $\boxed{K^{-1} = \frac{1}{|K|} adj(K)}$ → To get K^{-1}

$|K|$ = determinant of matrix 'd'

$d = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $d = [ad - bc]$

Ex: $d = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} = [12 - 9] = 3$

$d = 3$ determinant value

② Now, find multiplicative inverse of determinant 'd'

i.e. $d d^{-1} \equiv 1 \bmod 26$ [should be an identity matrix]

$3 * d^{-1} \equiv 1 \bmod 26$

$\Rightarrow (3 \times d^{-1}) \bmod 26 = 1$

$\boxed{d^{-1} = 9}$

[≡ →

- use trial method

$1 \bmod 26 = 1$

$(3 \times d^{-1}) \bmod 26 = 1$

$(3 * 9) \bmod 26 = 1$

$27 \bmod 26 = 1$

$\therefore d = 3, \quad d^{-1} = 9$

(3) Now we will find adjoint of the matrix

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then $adj(A) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

$K = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$, $adj(K) = \begin{bmatrix} 6 & -3 \\ -3 & 2 \end{bmatrix}$

Add 26 to remove -ve values before decryption process

$$adj(K) = \begin{bmatrix} 6 & -3+26 \\ -3+26 & 2 \end{bmatrix} = \begin{bmatrix} 6 & 23 \\ 23 & 2 \end{bmatrix}$$

$$adj(K) = \begin{bmatrix} 6 & 23 \\ 23 & 2 \end{bmatrix}$$

(4)

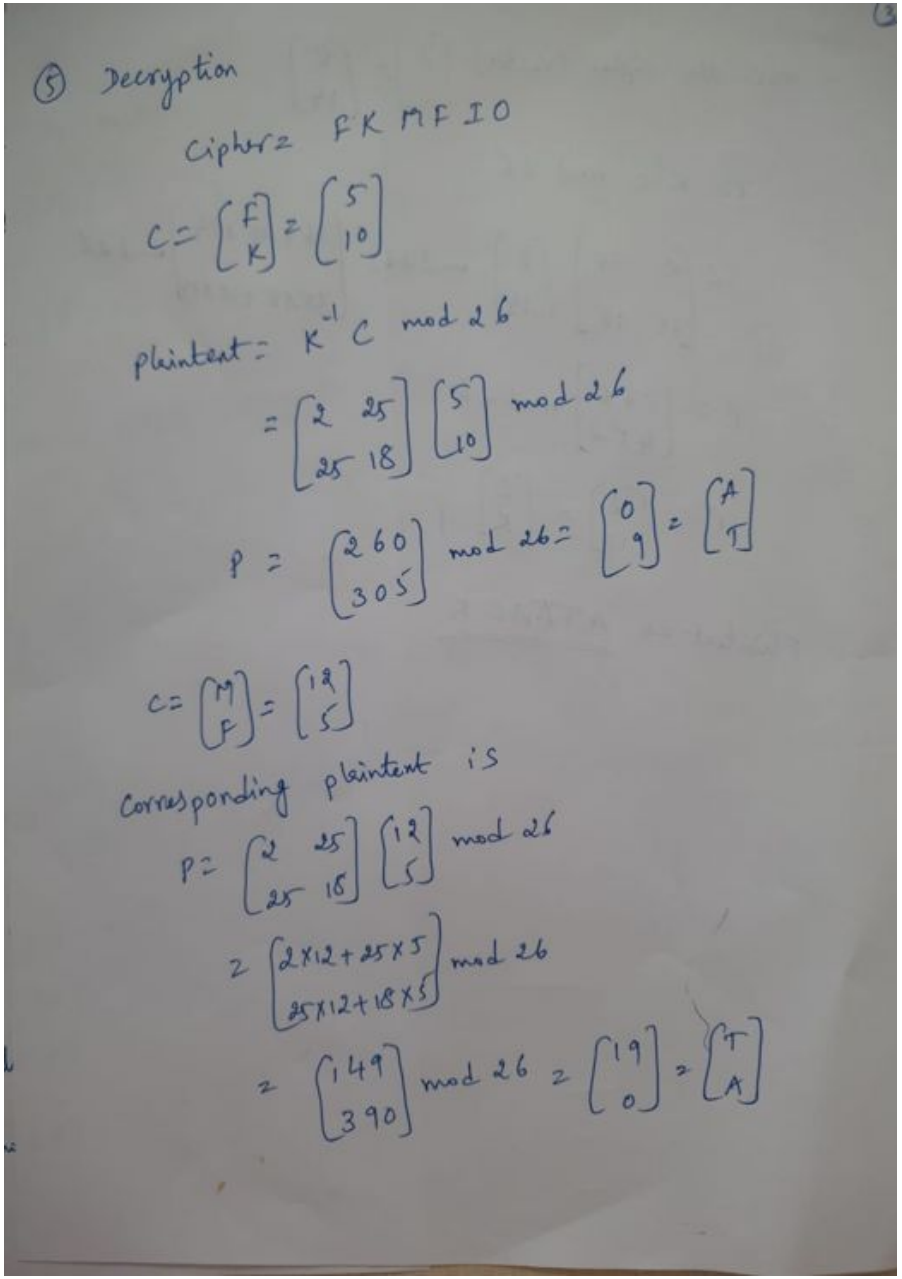
$$K^{-1} = \frac{1}{|K|} adj(K) = K^{-1} adj(K) = d^{-1} adj(K)$$

$$K^{-1} = 9 \begin{bmatrix} 6 & 23 \\ 23 & 2 \end{bmatrix} = \begin{bmatrix} 54 & 207 \\ 207 & 18 \end{bmatrix}$$

Now find its modulo 26

$$K^{-1} = \begin{bmatrix} 54 & 207 \\ 207 & 18 \end{bmatrix} \bmod 26 = \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix}$$

$$K^{-1} = \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix}$$



⑤ Decryption

Cipher = FKMFIO

$$C = \begin{bmatrix} F \\ K \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

plaintext = $K^{-1} C \bmod 26$

$$= \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix} \begin{bmatrix} 5 \\ 10 \end{bmatrix} \bmod 26$$

$$P = \begin{bmatrix} 260 \\ 305 \end{bmatrix} \bmod 26 = \begin{bmatrix} 0 \\ 9 \end{bmatrix} = \begin{bmatrix} A \\ J \end{bmatrix}$$

$$C = \begin{bmatrix} M \\ F \end{bmatrix} = \begin{bmatrix} 12 \\ 5 \end{bmatrix}$$

Corresponding plaintext is

$$P = \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix} \begin{bmatrix} 12 \\ 5 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 2\times12 + 25\times5 \\ 25\times12 + 18\times5 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 149 \\ 390 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 \\ 0 \end{bmatrix} = \begin{bmatrix} T \\ A \end{bmatrix}$$

TRANSPOSITION TECHNIQUES

- All the techniques examined so far involve the substitution of a cipher text symbol for a plaintext symbol. A very different kind of mapping is achieved by performing some sort of permutation on the plaintext letters. This technique is referred to as a transposition cipher.

- **Rail fence**
- is simplest of such cipher, in which the plaintext is written down as a sequence of diagonals and then read off as a sequence of rows.
- Plaintext = meet at the school house To encipher this message with a rail fence of depth 2, we write the message as follows:
- m e a t e c o l o s

  e t t h s h o h u e

  The encrypted message is MEATECOLOSETTHSHOHUE

- **Row Transposition Ciphers**
- A more complex scheme is to write the message in a rectangle, row by row, and read the message off, column by column, but permute the order of the columns. The order of columns then becomes the key of the algorithm. e.g., plaintext = meet at the school house

- Key = 4 3 1 2 5 6 7

PT = m e e t a t t
h e s c h o o
l h o u s e

CT = ESOTCUEEHMHLAHSTOETO

A pure transposition cipher is easily recognized because it has the same letter frequencies as the original plaintext. The transposition cipher can be made significantly more secure by performing more than one stage of transposition. The result is more complex permutation that is not easily reconstructed.

Simple Columnar Transposition Technique

The simple columnar transposition technique simply arranges the plain text as a sequence of rows of a rectangle that are read in columns randomly.

- **Basic Technique**

Variations of the basic transposition technique such as rail-fence technique exist

1. Write the plain-text message row by row in a rectangle of a pre-defined size.
2. Read the message column by column. However, it need not be in the order of columns 1, 2, 3, etc. It can be any random order such as 2, 3, 1, etc.
3. The message thus obtained is the cipher-text message.

Simple columnar transposition technique

Simple Columnar Transposition Technique

Let us examine the simple columnar transposition technique with an example.
Consider the same plaintext message 'Come home tomorrow'.
Let us understand how it can be transformed into cipher text using this technique.

Original plain-text message: ***Come home tomorrow***

1. Let us consider a rectangle with six columns. Therefore, when we write the message in the rectangle row by row (suppressing spaces), it would look as follows:

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6
C	o	m	e	h	o
m	e	t	o	m	o
r	r	o	w		

2. Now, let us decide the order of columns as some random order, say 4, 6, 1, 2, 5 and 3. Then read the text in the order of these columns.
3. The cipher text thus obtained would be ***eowoocmroerhmmto***.

Fig. 2.38 Example of simple columnar transposition technique

Simple Columnar Transposition Technique with Multiple Rounds

- To improve the basic simple columnar transposition technique, we can introduce more complexity.
- The idea is to use the same basic procedure as used by the simple columnar transposition technique, but to do it more than once. That adds considerably more complexity for the cryptanalyst.
- Cipher text produced by the simple columnar transposition technique with multiple rounds is much more complex to crack as compared to the basic technique.

1. Write the plain text message row by row in a rectangle of a pre-defined size.
2. Read the message column by column. However, it need not be in the order of columns 1, 2, 3 etc. It can be any random order such as 2, 3, 1, etc.
3. The message thus obtained is the cipher text message of round 1.
4. Repeat steps 1 to 3 as many times as desired.

Original plain-text message: ***Come home tomorrow***

1. Let us consider a rectangle with six columns. Therefore, when we write the message in the rectangle row by row, it would look as follows:

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6
C	o	m	e	h	o
m	e	t	o	m	o
r	r	o	w		

2. Now, let us decide the order of columns as some random order, say 4, 6, 1, 2, 5, 3. Then read the text in the order of these columns.
3. The cipher text thus obtained would be ***eowoocmroerhmmto*** in round 1.
4. Let us perform Steps 1 through 3 once more. So, the tabular representation of the cipher text after round 1 is as follows:

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6
e	o	w	o	o	c
m	r	o	e	r	h
m	m	t	o		

5. Now, let us use the same order of columns, as before, that is 4, 6, 1, 2, 5, and 3. Then read the text in the order of these columns.
6. The cipher text thus obtained would be ***oeochemmormorwot*** in round 2.
7. Continue like this if more number of iterations is desired, otherwise stop.

.2.40 Example of simple columnar transposition technique with multiple rounds

Book Cipher/Running-Key Cipher

- similar in principle to the Vernam cipher. For producing cipher text, some portion of text from a book is used, which serves the purpose of a one-time pad.
- Thus, the characters from a book are used as onetime pad, and they are added to the input plain-text message similar to the way a one-time pad works.

Steganography

- an alternative to encryption
- hides existence of message
  - using only a subset of letters/words in a longer message marked in some way
  - using invisible ink
  - hiding in LSB in graphic image or sound file
- has drawbacks
  - high overhead to hide relatively few info bits

- A plaintext message may be hidden in any one of the two ways. The methods of steganography conceal the existence of the message, whereas the methods of cryptography render the message unintelligible to outsiders by various transformations of the text.
- A simple form of steganography, but one that is time consuming to construct is one in which an arrangement of words or letters within an apparently innocuous text spells out the real message. e.g., (i) the sequence of first letters of each word of the overall message spells out the real (hidden) message. (ii) Subset of the words of the overall message is used to convey the hidden message. Various other techniques have been used historically, some of them are

Various other techniques have been used historically, some of them are

- Character marking – selected letters of printed or typewritten text are overwritten in pencil. The marks are ordinarily not visible unless the paper is held to an angle to bright light.
- Invisible ink – a number of substances can be used for writing but leave no visible trace until heat or some chemical is applied to the paper.
- Pin punctures – small pin punctures on selected letters are ordinarily not visible unless the paper is held in front of the light.
- Typewritten correction ribbon – used between the lines typed with a black ribbon, the results of typing with the correction tape are visible only under a strong light

Drawbacks of Steganography

- Requires a lot of overhead to hide a relatively few bits of information.
- Once the system is discovered, it becomes virtually worthless.

More Definitions

- **unconditional security**
  - no matter how much computer power is available, the cipher cannot be broken since the ciphertext provides insufficient information to uniquely determine the corresponding plaintext
- **computationally secure**
  - The cost of breaking the cipher exceeds the value of the encrypted information.
  - The time required to break the cipher exceeds the useful lifetime of the information.